

Welcome

CS 472/572
Unsupervised Machine Learning

Hierarchical Clustering

Week-6 -Class -1

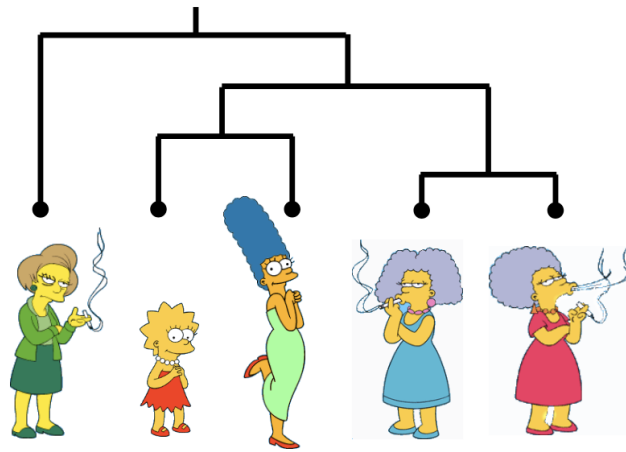
Spring 2026
Instructor: Dr. Khondker Fariha Hossain

Two Types of Clustering

- **Partitional algorithms:** Construct various partitions and then evaluate them by some criterion
- **Hierarchical algorithms:** Create a hierarchical decomposition of the set of objects using some criterion (focus of this class)

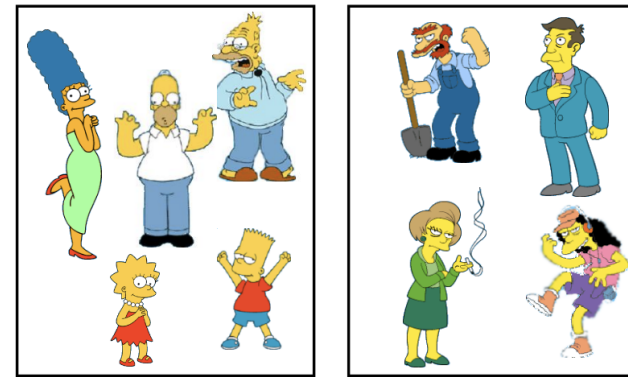
Bottom up or top down

Hierarchical



Top down

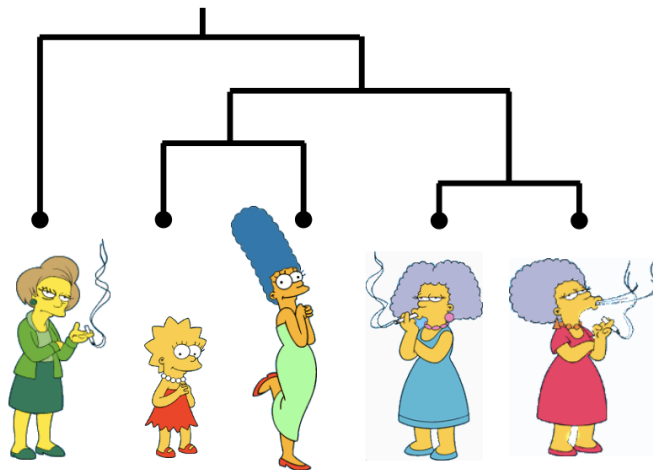
Partitional



(How-to) Hierarchical Clustering

The number of dendrograms with n leaves = $(2n - 3)! / [(2^{n-2}) (n - 2)!]$

Number of Leafs	Number of Possible Dendrograms
2	1
3	3
4	15
5	105
...	...
10	34,459,425



Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

(How-to) Hierarchical Clustering

We begin with a distance matrix which contains the distances between every pair of objects in our database.

$$D(\text{Mrs. Krabappel}, \text{Lisa Simpson}) = 8$$

$$D(\text{Marge Simpson}, \text{Bart Simpson}) = 1$$

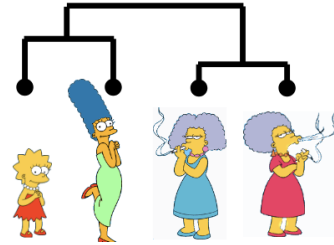
				
0	8	8	7	7
	0	2	4	4
		0	3	3
			0	1
				0

(How-to) Hierarchical Clustering

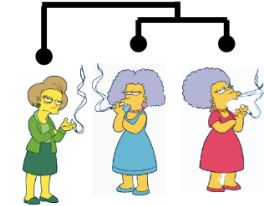
Bottom-Up (agglomerative):

Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

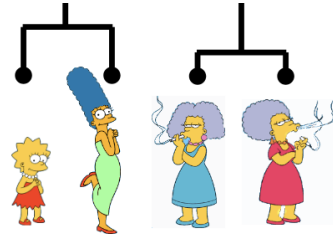
Consider all possible merges...



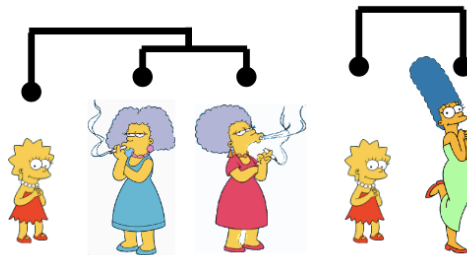
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Choose the best



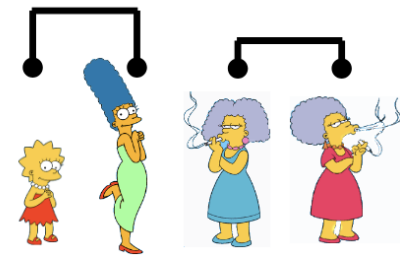
Consider all possible merges...



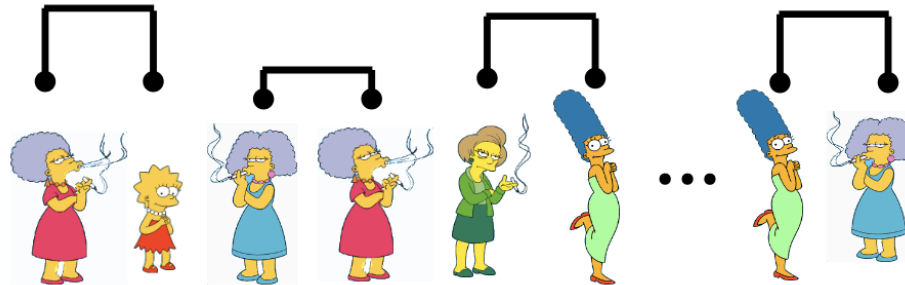
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Choose the best



Consider all possible merges...



...



Choose the best



Technique Characteristics

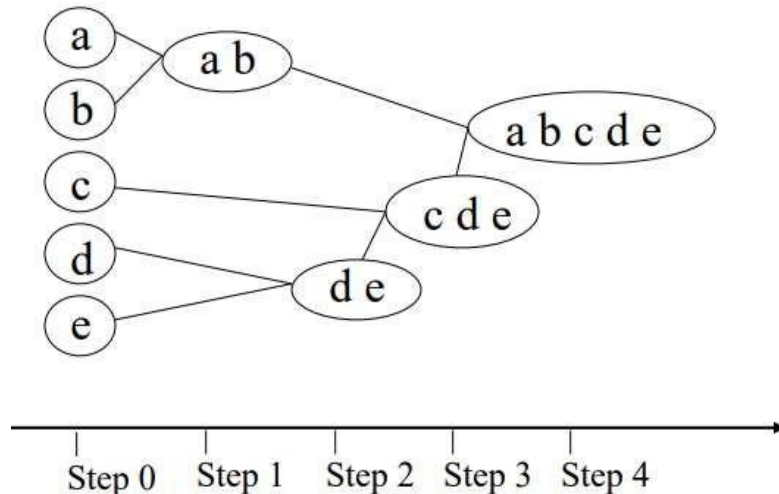
- Hard vs Fuzzy
 - Hard clustering assigns each instance to one cluster
 - fuzzy clustering assigns degree of membership
- Monothetic vs Polythetic
 - *Polythetic*: all attributes are used simultaneously, e.g., to calculate distance (most algorithms)
 - *Monothetic*: attributes are considered one at a time
- Incremental vs Non-Incremental
 - With large data sets it may be necessary to consider only part of the data at a time (data mining)
 - Incremental works instance by instance

Hierarchical Clustering

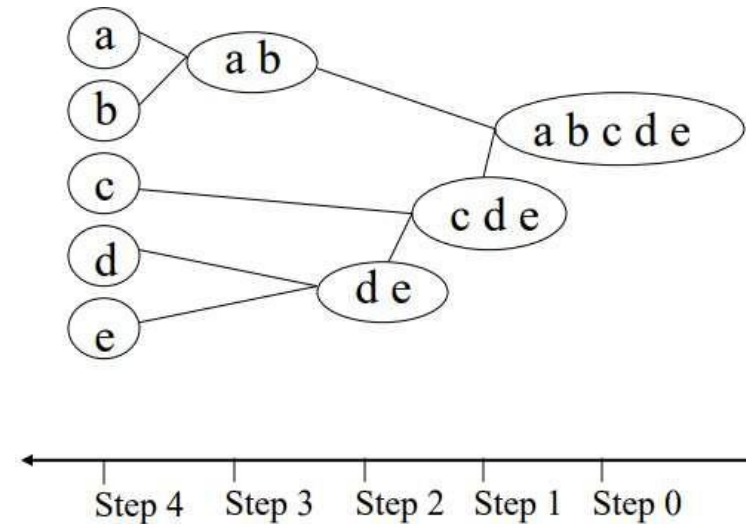
A hierarchical clustering is a set of nested clusters that are organized as a tree.

- Agglomerative vs Divisive
 - *Agglomerative*: each instance is its own cluster and the algorithm merges clusters
 - *Divisive*: begins with all instances in one cluster and divides it up

Agglomerative

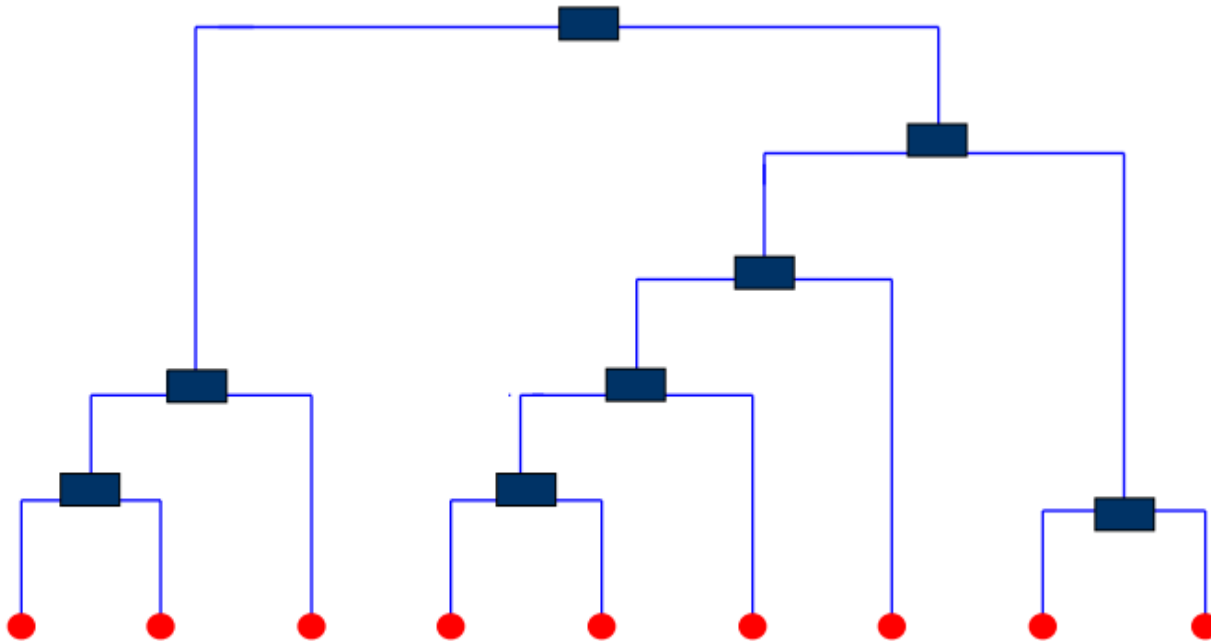


Divisive



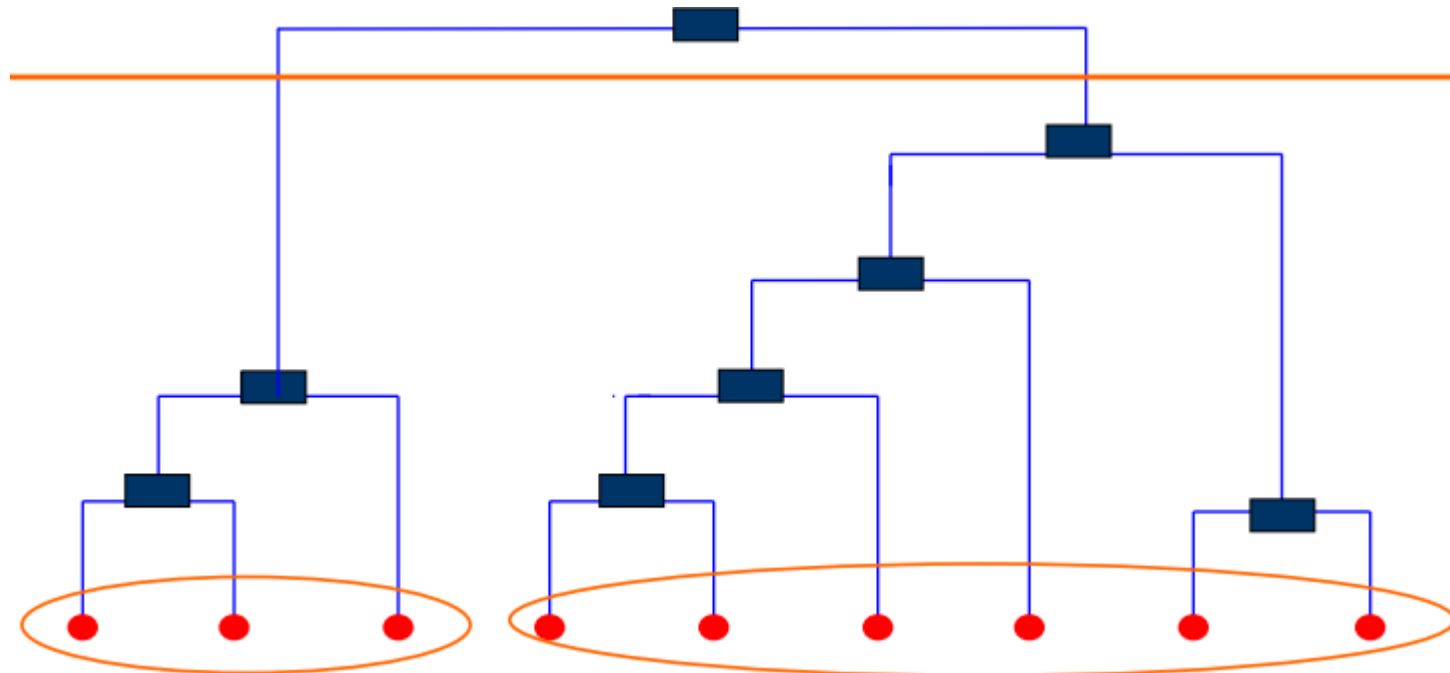
Dendrogram

- A tree that shows how clusters are merged/split hierarchically
- Each node on the tree is a cluster; each leaf node is a singleton cluster



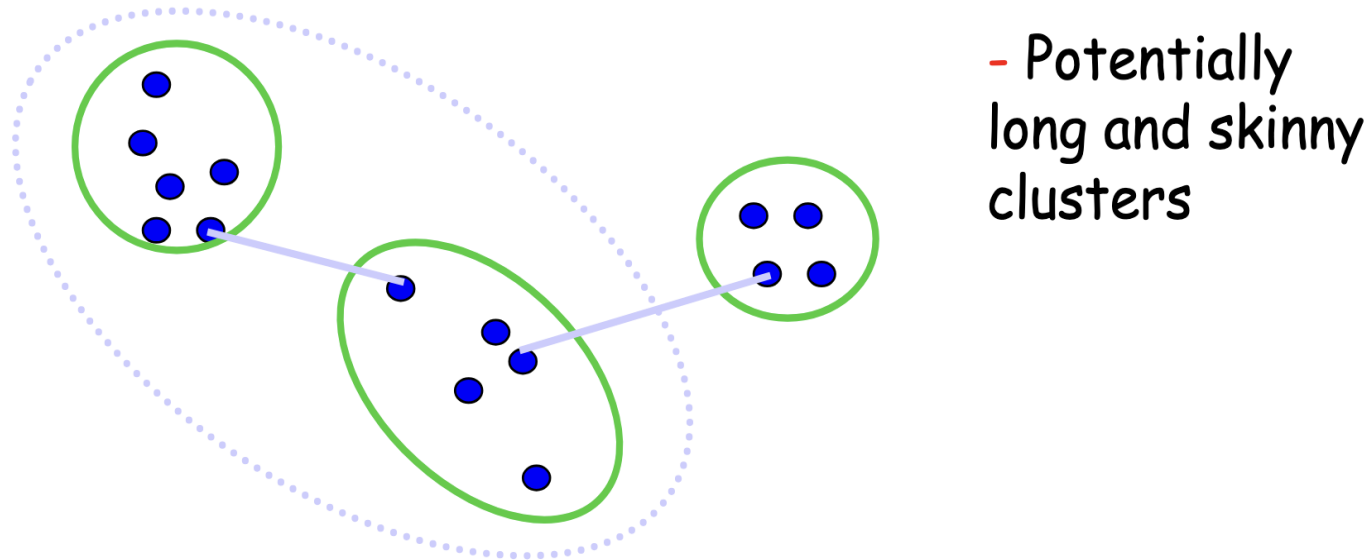
Dendrogram

- A clustering of the data objects is obtained by cutting the dendrogram at the desired level, then each connected component forms a cluster



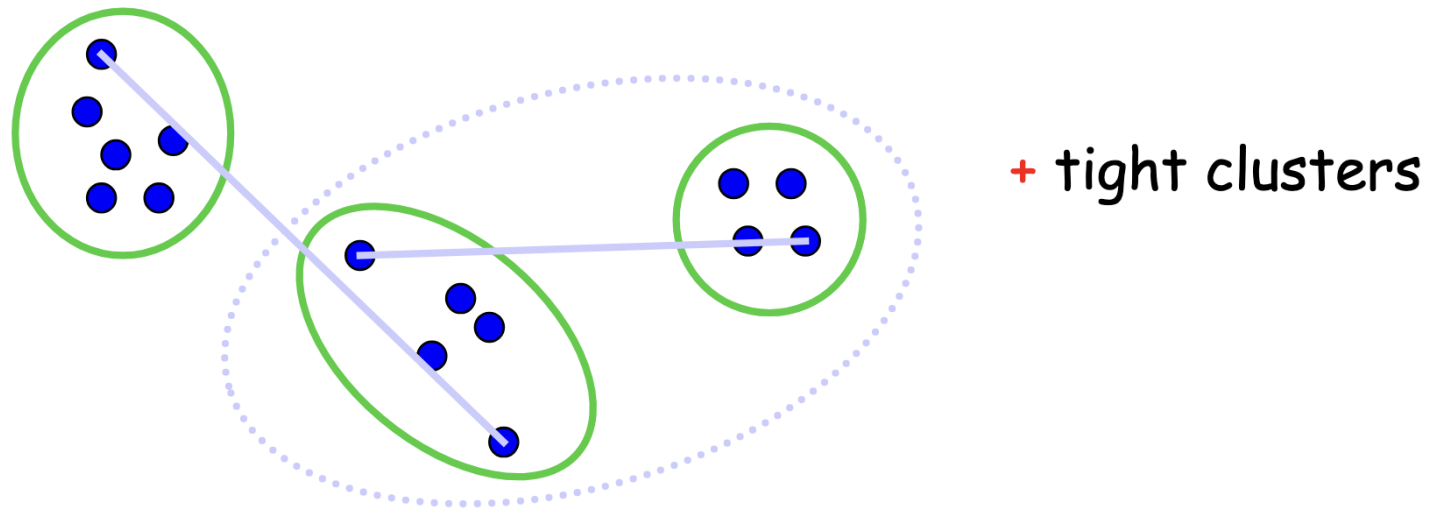
Computing distance between clusters: Single Link

- cluster distance = distance of two **closest** members in each class



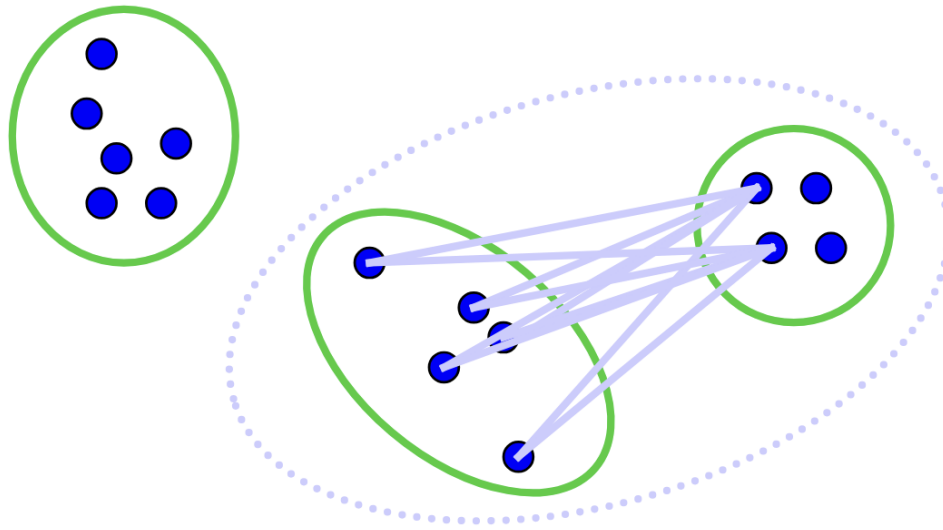
Computing distance between clusters: : Complete Link

- cluster distance = distance of two farthest members



Computing distance between clusters: Average Link

- cluster distance = **average distance** of all pairs

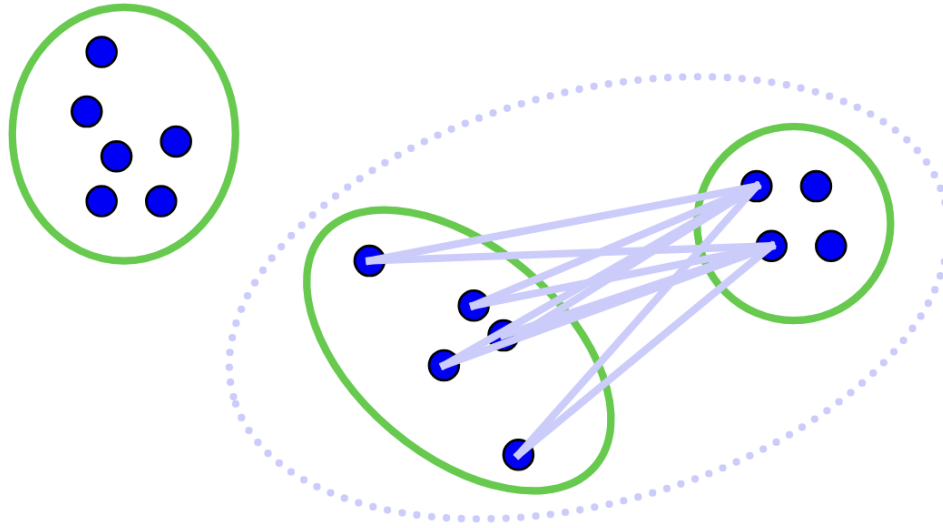


**the most widely
used measure**

**Robust against
noise**

Computing distance between clusters: Average Link

- cluster distance = **average distance** of all pairs



**the most widely
used measure**

**Robust against
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Computing distance between clusters: Average Link

Single Linkage: distance between closest points

Complete Linkage: distance between farthest points

Average Linkage: mean pairwise distance across all points in the two clusters

Computing distance between clusters: Average Link

- ◆ Single Linkage (Nearest-Neighbor Linkage)

What it does:

Measures distance between the closest points in two clusters

Tends to create long, chain-like clusters (a phenomenon called chaining)

Very sensitive to noise and outliers

Use When:

You want to detect non-spherical, elongated structures

You care about preserving connectivity even when clusters are oddly shaped

Real-Life Example:

Detecting road or river networks in satellite images: Single linkage can follow meandering paths and connect points that form long, skinny shapes (e.g., a highway or river).

Computing distance between clusters: Average Link

Complete Linkage (Farthest-Neighbor Linkage):

What it does:

Measures distance between the farthest points in two clusters

Creates compact, well-separated, tight clusters Less sensitive to noise, but may break long clusters apart

Use When:

You want compact clusters with clear boundaries

You need to avoid very spread-out or elongated groups

Real-Life Example:

Grouping customers for premium-targeting in marketing

Complete linkage ensures each customer in a cluster is similar to all others, giving you high-quality, tightly defined segments.

Computing distance between clusters: Average Link

Average Linkage (UPGMA: Unweighted Pair Group Method with Arithmetic Mean)

What it does:

- Computes the average pairwise distance between all points in two clusters
- Provides a balanced compromise between single and complete linkage
- Less sensitive to noise than single linkage, less strict than complete linkage

Use When:

- You want stable, general-purpose clustering
- You don't need extreme compactness or sensitivity
- Data has moderately well-defined, but not perfectly spherical clusters

Real-Life Example:

Clustering documents or articles by topic. Average linkage handles varied writing styles and vocabulary while still grouping documents by general similarity.

Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward
 1. Compute the distance matrix
 2. Let each data point be a cluster
 3. Repeat
 4. Merge the two closest clusters
 5. Update the distance matrix
 6. Until only a single cluster remains
- Key operation is the computation of the distance between two clusters
- Different approaches to defining the distance between clusters distinguish the different algorithms

Agglomerative Clustering Algorithm

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Agglomerative Hierarchical Clustering - Single Linkage

Agglomerative Hierarchical Clustering - Single

- Consider the following set of 6 one dimensional data points:
- 18, 22, 25, 42, 27, 43
- Apply the **agglomerative hierarchical clustering** algorithm to build the hierarchical clustering **dendogram**.
- Merge the clusters using **Min distance** and update the proximity matrix accordingly.
- Clearly show the **proximity matrix** corresponding to each iteration of the algorithm.

Agglomerative Hierarchical Clustering - Single

Step – 1

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

Merge: (42, 43)

Agglomerative Hierarchical Clustering - Single

Step – 2

	18	22	25	27	42, 43
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	2	17
27	9	5	2	0	15
42, 43	24	20	17	15	0

Merge: (42,43)(25, 27)

Agglomerative Hierarchical Clustering - Single

Step – 3

	18	22	25, 27	42, 43
18	0	4	7	24
22	4	0	3	20
25, 27	7	3	0	15
42, 43	24	20	15	0

Merge: ((42,43)(25, 27), 22)

Agglomerative Hierarchical Clustering - Single

Step – 4

	18	22, 25, 27	42, 43
18	0	4	24
22, 25, 27	4	0	15
42, 43	24	15	0

Agglomerative Hierarchical Clustering - Single

Step – 5

	18, 22, 25, 27	42, 43
18, 22, 25, 27	0	15
42, 43	15	0

Merge: ((18,22,25,27), (42,43))

Agglomerative Hierarchical Clustering - Single

Step – 6

	18, 22, 25, 27, 42, 43
18, 22, 25, 27, 42, 43	0

Agglomerative Hierarchical Clustering - Single

Step – 5

	18, 22, 25, 27,	42, 43
18, 22, 25, 27	0	15
42, 43	15	0

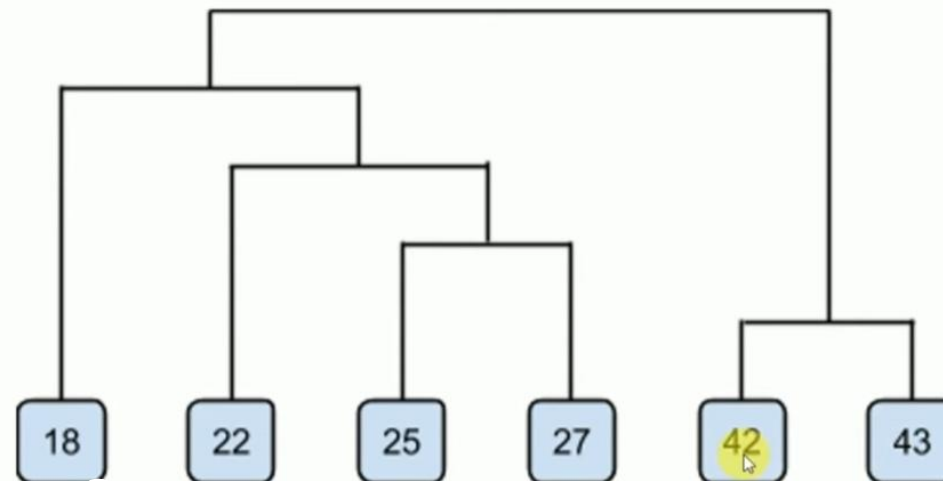
Merge: ((42,43),((25,27),22), 18)

Agglomerative Hierarchical Clustering - Single

Agglomerative Hierarchical Clustering Solved Example

- Dendrogram

$((42, 43), ((25, 27), 22), 18)$



Agglomerative Hierarchical Clustering

– Complete Linkage

Agglomerative Hierarchical Clustering – Complete Linkage

- Given a one dimensional Dataset: {1, 5, 8, 10, 2}, use the agglomerative clustering with the complete link with Euclidian distance to establish a hierarchical grouping relationship.

By using the cutting threshold of 5, how many clusters are there?

Agglomerative Hierarchical Clustering – Complete Linkage

$$\text{Euclidean distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Euclidean distance} = \sqrt{(x_2 - x_1)^2}$$

- In order to use the agglomerative algorithm,
- we need to calculate the distance matrix.
- One-dimensional data set {1, 5, 8, 10, 2}

	1	5	8	10	2
1	0	4	7	9	1
5	4	0	3	5	3
8	7	3	0	2	6
10	9	5	2	0	8
2	1	3	6	8	0

Agglomerative Hierarchical Clustering – Complete Linkage

- $d(2, \{1,5\}) = \max\{d(2,1), d(2,5)\} = \max\{4, 3\} = 4$
- $d(3, \{1,5\}) = \max\{d(3,1), d(3,5)\} = \max\{7, 6\} = 7$
- $d(4, \{1,5\}) = \max\{d(4,1), d(4,5)\} = \max\{9, 8\} = 9$

• Let the 1st column (row) denote the distances between this cluster and other points, we have the following distance matrix:

	1	2	3	4	5
1	0	4	7	9	1
2	4	0	3	5	3
3	7	3	0	2	6
4	9	5	2	0	8
5	1	3	6	8	0

	1,5	2	3	4
1,5	0	4	7	9
2	4	0	3	5
3	7	3	0	2
4	9	5	2	0

Agglomerative Hierarchical Clustering – Complete Linkage

- From the above distance matrix, we can see the distance between points 3 and 4 is smallest.

- Hence, they merge together to form a cluster {3, 4}.

- Using the complete link, we have the distance between different points/clusters as follows:

- $d(\{1,5\}, \{3,4\}) = \max\{d(\{1,5\}, 3), d(\{1,5\}, 4)\} = \max\{7, 9\} = 9$

- $d(2, \{3,4\}) = \max\{d(2,3), d(2,4)\} = \max\{3, 5\} = 5$

- Thus, we can update the distance matrix, where row 2 corresponds to point 2, rows 1 and 3 correspond to clusters {1,5} and {3,4}, as follows:

	1,5	2	3	4
1,5	0	4	7	9
2	4	0	3	5
3	7	3	0	2
4	9	5	2	0

	1,5	2	3,4
1,5	0	4	9
2	4	0	5
3,4	9	5	0

Agglomerative Hierarchical Clustering – Complete Linkage

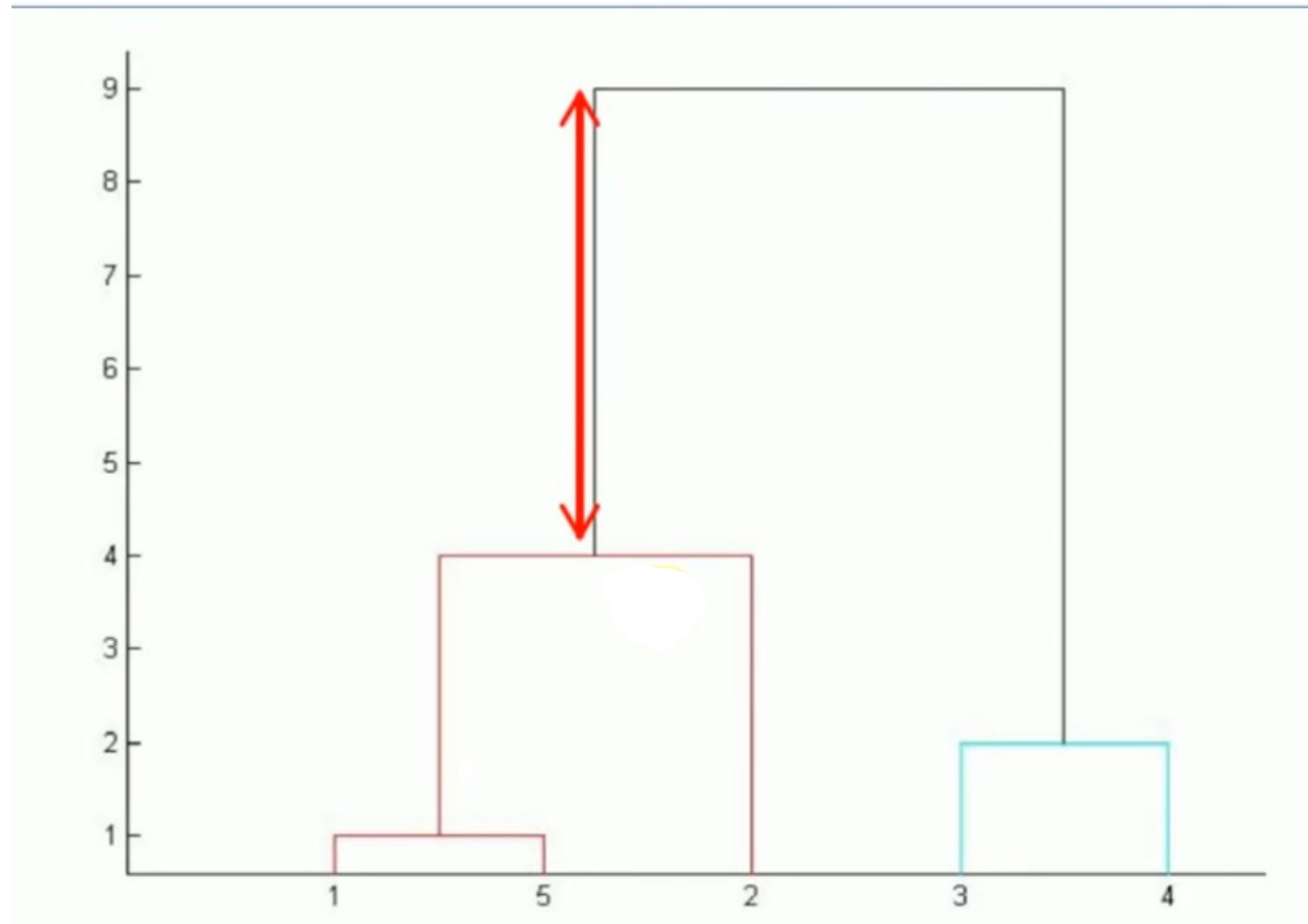
Following the same procedure, we merge point 2 with the cluster {1, 5} to form {1,2,5} and update the distance matrix as follows:.

	[1,5], 2	[3,4]
[1,5], 2	0	9
[3,4]	9	0

	1,5	2	3,4
1,5	0	4	9
2	4	0	5
3,4	9	5	0

After increasing the distance threshold to 9, all clusters would merge

Agglomerative Hierarchical Clustering – Complete Linkage



Thank You

Reference:

CMU : 10601- Machine learning
Hierarchical Clustering – Mahesh Huddar